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Week 1 – Introduction

Study Material

Week I

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1. Introduction to Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that focuses on enabling systems to learn and make decisions or predictions without being explicitly programmed. It involves the development of algorithms and statistical models that allow computers to identify patterns in data, make decisions or perform tasks based on input data.

The concept of ML revolves around three key components:

- **Data**: The foundational element of machine learning. It can be structured (tables, databases) or unstructured (images, text, videos).
- **Algorithms**: The mathematical methods used to process data and learn patterns.
- **Model**: The result of applying algorithms to data to make predictions or decisions.

Machine learning has found applications in various domains, including healthcare, finance, marketing, robotics and natural language processing.

1.1 Types of Machine Learning

Machine learning can be broadly classified into three main types based on the nature of the learning process and the type of data available.

1.1.1 Supervised Learning

In supervised learning, the model is trained on a labelled dataset, where both input (features) and output (labels or targets) are known. The goal is to learn a mapping function from inputs to outputs and use it to make predictions on new, unseen data.

- **Key Features**:
  - Requires labelled data.
  - The training data includes both inputs and their corresponding correct outputs.
  - Focuses on minimising the error between predicted and actual outputs.
- **Examples**:
  - **Regression**: Predicting a continuous value (e.g., house price prediction, stock market forecasting).
  - **Classification**: Predicting discrete categories (e.g., spam email detection, disease diagnosis).
- **Common Algorithms**:
  - Linear Regression
  - Logistic Regression
  - Support Vector Machines (SVM)
  - Decision Trees

  - Random Forests
  - Neural Networks
- Key Components of Supervised Learning

- **Input (Features)**:
  - These are the variables used to make predictions.
  - Example: In a house price prediction model, inputs could include the size of the house, the number of bedrooms and the location.
- **Output (Labels)**:
  - The known outcomes or target values the model aims to predict.
  - Example: The house price in the previous scenario.
- **Training Data**:
  - A collection of input-output pairs used to train the model.
  - Example: Historical data where house features and their corresponding prices are known.
- **Model**:
  - A mathematical function or algorithm that maps inputs to outputs.
  - Example: $y=f(x)$, where x is the input, and y is the output.
- **Loss Function**:
  - Measures the difference between the predicted output and the actual output.
  - Example: Mean Squared Error (MSE) in regression or Cross-Entropy Loss in classification.
- **Optimisation Algorithm**:
  - Adjusts the model parameters to minimise the loss function.
  - Example: Gradient Descent.

1.1.2 Unsupervised Learning

Unsupervised learning deals with unlabelled data, where the system tries to learn patterns and structures from the input data without predefined labels.

- **Key Features**:
  - Does not require labelled data.
  - Focuses on discovering hidden patterns, structures or relationships within the data.
  - Commonly used for clustering, dimensionality reduction, and association rule mining.
- **Examples**:
  - **Clustering**: Grouping similar data points (e.g., customer segmentation, image segmentation).
  - **Dimensionality Reduction**: Reducing the number of features while preserving meaningful information (e.g., Principal Component Analysis (PCA)).

- **Common Algorithms**:
  - K-Means Clustering
  - Hierarchical Clustering
  - DBSCAN (Density-Based Spatial Clustering)
  - Autoencoders
  - PCA

- Key Components of Unsupervised Learning

Unsupervised learning is a type of machine learning where the algorithm works with unlabelled data to find patterns, structures or relationships within the data. It aims to explore the data without predefined labels or outcomes.

Below are the key components that define unsupervised learning:

- Input Data (Features)

- **Definition**: The set of variables or attributes describing the data.
- **Nature**: Unlike supervised learning, the data is unlabelled, meaning there are no corresponding target outputs.
- **Example**:
  - Customer demographic data for segmentation (e.g., age, income, spending score).
  - Pixel intensity values in an image for clustering similar objects.

- Output (Patterns or Structures)

- **Goal**: Instead of predicting labels or values, the focus is on discovering hidden structures in the data.
- **Outputs in Unsupervised Learning**:
  - **Clusters**: Grouping similar data points (e.g., customers with similar shopping habits).
  - **Anomalies**: Identifying data points that deviate significantly from the norm (e.g., fraud detection).
  - **Reduced Dimensions**: Simplifying data representation by retaining important features (e.g., principal components in PCA).

- Objective Function

- **Definition**: A function that evaluates how well the model identifies patterns or structures.
- **Examples**:
  - **Clustering**: Minimise within-cluster variance to ensure data points in the same cluster are similar.

    - **Dimensionality Reduction**: Minimise reconstruction error when representing high-dimensional data in lower dimensions.

- Model or Algorithm

- **Definition**: The mathematical or computational approach used to analyse and structure the data.
- **Common Algorithms**:
    - **Clustering Algorithms**:
        - K-Means.
        - Hierarchical Clustering.
        - DBSCAN (Density-Based Spatial Clustering).
    - **Dimensionality Reduction Techniques**:
        - Principal Component Analysis (PCA).
        - t-SNE (t-Distributed Stochastic Neighbour Embedding).
        - Autoencoders.
    - **Anomaly Detection**:
        - Isolation Forest.
        - Gaussian Mixture Models (GMM).

- Distance or Similarity Metrics

- **Definition**: Measures used to assess how similar or different data points are.
- **Examples**:
    - **Euclidean Distance**: Commonly used in clustering tasks (e.g., K-Means).
    - **Cosine Similarity**: Used for text and document similarity.
    - **Manhattan Distance**: Measures distance along axes at right angles.

- Learning Paradigm

- **Exploration-Driven**:
    - No predefined outputs; the algorithm must infer structure from the input data.
    - Focus is on discovering natural groupings, correlations or significant features.
- **Iterative Process**:
    - Models often refine their outputs iteratively (e.g., K-Means adjusts centroids in successive steps).

- Validation Metrics

- **Challenge**: No labelled data means traditional accuracy metrics cannot be used.
- **Validation Methods**:
    - **Silhouette Score**: Evaluates the consistency of clustering.
    - **Elbow Method**: Identifies the optimal number of clusters in K-Means.
    - **Reconstruction Error**: Measures how well dimensionality reduction techniques preserve original data.

- Hyperparameters

- **Definition**: Parameters that control the learning process and model behaviour.
- **Examples**:
  - **K (Number of Clusters)**: For algorithms like K-Means.
  - **Epsilon (ϵ\epsilonϵ) and MinPoints**: For DBSCAN, defining the density threshold for clusters.
  - **Number of Components**: In PCA, determines how many principal components to retain.

- Data Pre-processing

- **Importance**: Ensures the input data is suitable for analysis and enhances algorithm performance.
- **Techniques**:
  - **Normalisation**: Scales features to a common range.
  - **Standardisation**: Transforms data to have zero mean and unit variance.
  - **Handling Missing Data**: Imputation techniques to fill missing values.

- Real-World Applications

- **Clustering**:
  - Customer segmentation in marketing.
  - Grouping similar documents in information retrieval.
- **Dimensionality Reduction**:
  - Data visualisation (e.g., reducing high-dimensional data to 2D or 3D plots).
  - Pre-processing for supervised learning to improve efficiency.
- **Anomaly Detection**:
  - Fraud detection in banking.
  - Identifying faulty equipment in manufacturing.

1.1.3 Reinforcement Learning

In reinforcement learning, an agent learns by interacting with an environment. The agent performs actions to maximise a reward signal while navigating through the environment. This type of learning is inspired by behavioural psychology.

- **Key Features**:
  - Involves an agent, environment, actions, and rewards.
  - No explicit supervision; learning happens through trial and error.
  - Used for sequential decision-making problems.
- **Examples**:
  - Robotics (e.g., teaching robots to walk or pick objects).

  - Game playing (e.g., AlphaGo mastering the game of Go).
  - Autonomous driving.
- **Common Algorithms**:
  - Q-Learning
  - Deep Q-Networks (DQN)
  - Policy Gradient Methods
  - Actor-Critic Methods

1.1.4 Other Emerging Types of Learning

- Semi-Supervised Learning

- Combines a small amount of labelled data with a large amount of unlabelled data.
- Useful when labelling data is expensive or time-consuming.
- Examples: Text classification, medical image analysis.

- Self-Supervised Learning

- The model generates its labels from the input data itself.
- Used in fields like computer vision and natural language processing (e.g., BERT for NLP).

2. Machine Learning Model Development

Developing a machine learning model involves a series of structured steps to ensure the model is accurate, efficient and effective. Here's a detailed explanation of the key steps involved:

2.1 Problem Definition

- **Objective**: Clearly define the problem you want to solve and the goals of the machine learning model.
- **Tasks**:
  - Identify the type of machine learning (e.g., classification, regression, clustering, reinforcement).
  - Determine the expected output (e.g., predictions, insights).
  - Understand the domain and constraints.

2.2 Data Collection

- **Objective**: Gather relevant data to train and test the model.

- **Tasks**:
    - Collect data from various sources (e.g., databases, APIs, sensors, web scraping).
    - Ensure the data is relevant, sufficient and representative of the problem.

2.3 Data Preprocessing

- **Objective**: Prepare the data for analysis by cleaning and formatting it.
- **Tasks**:
    - Handle missing values (e.g., imputation or removal).
    - Remove duplicates and outliers.
    - Normalise or standardise the data if required.
    - Convert categorical variables into numerical representations (e.g., one-hot encoding).
    - Split the data into training, validation, and testing sets (e.g., 70%-15%-15%).

2.4 Exploratory Data Analysis (EDA)

- **Objective**: Understand the data distribution, relationships, and patterns.
- **Tasks**:
    - Visualise data using plots (e.g., histograms, scatter plots, heatmaps).
    - Identify correlations between features.
    - Analyse feature importance and distributions.
    - Detect anomalies or biases in the data.

2.5 Feature Engineering

- **Objective**: Enhance the dataset by creating, selecting, or transforming features.
- **Tasks**:
    - Feature Selection: Choose the most relevant features.
    - Feature Creation: Generate new features from existing ones (e.g., polynomial features).
    - Feature Transformation: Apply techniques like scaling, normalisation or log transformation.

2.6 Model Selection

- **Objective**: Choose the appropriate algorithm(s) for the problem.
- **Tasks**:
    - Consider the nature of the problem (e.g., regression, classification).

- Select a few candidate algorithms for comparison (e.g., Random Forest, SVM, Neural Networks).
- Factor in computational resources and the size of the dataset.

2.7 Model Training

- **Objective**: Train the model on the training dataset.
- **Tasks**:
  - Define the algorithm and its hyperparameters.
  - Fit the model to the training data.
  - Evaluate performance on the training set (e.g., accuracy, loss).

2.8 Model Evaluation

- **Objective**: Assess the model's performance on unseen data.
- **Tasks**:
  - Test the model on the validation or testing set.
  - Use performance metrics suitable for the task:
    - Classification: Accuracy, Precision, Recall, F1 Score, ROC-AUC.
    - Regression: Mean Absolute Error (MAE), Mean Squared Error (MSE), R-squared.
  - Perform cross-validation to assess generalisation.

2.9 Hyperparameter Tuning

- **Objective**: Optimise the model's hyperparameters to improve performance.
- **Tasks**:
  - Use techniques like Grid Search, Random Search or Bayesian Optimisation.
  - Evaluate models with tuned hyperparameters on the validation set.

2.10 Model Deployment

- **Objective**: Integrate the trained model into a production environment.
- **Tasks**:
  - Save the model (e.g., pickle, joblib).
  - Deploy using APIs, cloud platforms or edge devices.
  - Ensure scalability and reliability.

2.11 Monitoring and Maintenance

- **Objective**: Ensure the model continues to perform well over time.

- **Tasks**:
  - Monitor the model for data drift or performance degradation.
  - Update or retrain the model with new data.
  - Log predictions and errors for future analysis.

2.12 Documentation and Communication

- **Objective**: Document the entire process and communicate findings.
- **Tasks**:
  - Record the assumptions, methods and results.
  - Share insights with stakeholders.
  - Ensure reproducibility by maintaining version control for data, code and models.

3. Supervised and Unsupervised Learning

Supervised and unsupervised learning are two primary categories of machine learning techniques. They differ in how they learn from data, their objectives, and their applications. Here's a detailed comparison:

3.1 Definition

- **Supervised Learning**:
  - Involves learning from labelled data, where the input features (X) and the corresponding output labels (Y) are provided. The goal is to learn a mapping function to predict the output for new, unseen data.
- **Unsupervised Learning**:
  - Involves learning from unlabelled data, where only input features (X) are available. The goal is to discover hidden patterns, structures or relationships within the data.

3.2 Data Requirements

- **Supervised Learning**:
  - Requires labelled data, which can be costly and time-consuming to obtain.
  - Example: For a spam email classifier, each email must be labelled as "spam" or "not spam."
- **Unsupervised Learning**:
  - Does not require labelled data, making it more flexible and scalable.

- Example: Customer segmentation, where customer purchase data is analysed without predefined group labels.

3.3 Learning Process

- **Supervised Learning**:
  - The algorithm learns by minimising the error between the predicted and actual outputs.
  - It uses feedback in the form of correct labels to improve the model during training.
- **Unsupervised Learning**:
  - The algorithm learns by finding patterns, clusters or associations in the data.
  - There is no explicit feedback or correct output to guide the learning process.

3.4 Objectives

- **Supervised Learning**:
  - Predict the output for new inputs.
  - Example Tasks:
    - **Classification**: Assign input data to predefined categories (e.g., spam vs. not spam).
    - **Regression**: Predict continuous values (e.g., house prices, temperature).
- **Unsupervised Learning**:
  - Understand the structure and distribution of data.
  - Example Tasks:
    - **Clustering**: Group similar data points (e.g., customer segmentation).
    - **Dimensionality Reduction**: Reduce the number of features while retaining important information (e.g., Principal Component Analysis).

3.5 Output

- **Supervised Learning**:
  - The output is a labelled result, such as a predicted category or value.
  - Example: Predicting whether a loan applicant will default (Yes/No).
- **Unsupervised Learning**:
  - The output is a pattern, cluster, or reduced feature set, not a specific label.
  - Example: Grouping customers into clusters based on purchasing behaviour.

3.6 Applications

- **Supervised Learning**:
  - Fraud Detection: Identifying fraudulent transactions.
  - Medical Diagnosis: Predicting diseases based on symptoms or test results.
  - Stock Price Prediction: Forecasting future prices based on historical data.
  - Sentiment Analysis: Analysing customer reviews to classify sentiment.
- **Unsupervised Learning**:
  - Market Segmentation: Identifying groups of similar customers for targeted marketing.
  - Anomaly Detection: Finding outliers in manufacturing or network traffic.
  - Recommendation Systems: Suggesting products by finding patterns in user behaviour.
  - Gene Expression Analysis: Grouping similar genes based on expression levels.

3.7 Complexity and Scalability

- **Supervised Learning**:
  - Relatively simpler to train and evaluate due to the availability of labelled data.
  - Scalability depends on the size of the labelled dataset.
- **Unsupervised Learning**:
  - More challenging to evaluate because there are no labels to validate the results.
  - Can handle large-scale datasets but might struggle to interpret results.

3.8 Examples

- **Supervised Learning**:
  - Predicting house prices based on features like size, location and number of rooms.
  - Classifying images of animals into categories (e.g., cat, dog, bird).
- **Unsupervised Learning**:
  - Grouping customers into clusters based on their buying habits.
  - Reducing the dimensions of a dataset with many features for visualisation.

3.9 Summary Table

Aspect	Supervised Learning	Unsupervised Learning
Data Requirement	Labelled data	Unlabelled data

Aspect	Supervised Learning	Unsupervised Learning
Objective	Predict output	Find patterns
Algorithms	Linear Regression, SVM, Neural Networks	K-Means, PCA, Auto encoders
Output	Labelled predictions	Clusters or reduced features
Applications	Fraud detection, stock prediction	Market segmentation, anomaly detection
Evaluation	Easier to validate	Harder to interpret and validate

4. Formal Model in Machine Learning

A **formal model** provides a mathematical framework to define and analyse machine learning processes. It specifies the structure, assumptions and properties of the learning problem and algorithm.

4.1 Key Components of the Formal Model

- **Instance Space (X):**
  - The set of all possible inputs (e.g., feature vectors).
  - Example: For image classification, {X} could be all possible pixel arrays.
- **Label Space (Y)**:
  - The set of all possible outputs or labels.
  - Example: Y={0,1} for binary classification or Y=R for regression tasks.
- **Hypothesis Space (H)**:
  - The set of all possible functions or models the learning algorithm can consider.
  - Example: Linear classifiers, decision trees, or neural networks.
- **Target Function (f)**:
  - The unknown function that maps inputs to outputs, $f:X\rightarrow Y$.
  - The goal of learning is to approximate ff using data.
- **Training Data (SS)**:
  - A finite sample of examples drawn from the instance space and labelled by f: $S=\{(x_1,y_1),\ldots,(x_m,y_m)\}$, where $y_i=f(x_i)$
- **Learning Algorithm (AA)**:
  - The procedure that selects a hypothesis $h \in H$ based on the training data.
  - Example: Gradient Descent, Decision Tree Construction.
- **Loss Function (L(h(x),y)L(h(x), y))**:

- A measure of the error between the predicted output h(x) and the true output y.
- Example: Mean Squared Error for regression, Cross-Entropy Loss for classification.

- **Generalisation**:
  - The ability of the learned model to perform well on unseen data, not just the training set.

5. PAC Learning (Probably Approximately Correct Learning)

PAC Learning is a theoretical framework introduced by Leslie Valiant in 1984 to study the feasibility of learning in the presence of uncertainty.

5.1 Core Concepts of PAC Learning

- **Instance Space (X)**:
  - A finite or infinite set of all possible inputs.
- **Target Concept (c)**:
  - The true function to be learned, c∈Cc \in \mathcal{C}, where C\mathcal{C} is the concept class (a set of possible target functions).
- **Hypothesis Class (H)**:
  - The set of all functions the learning algorithm can choose from to approximate cc.
- **Distribution (DD)**:
  - A probability distribution over the instance space X, representing how data is sampled.
- **Training Sample (SS)**:
  - A set of labelled examples $\{(x_1,c(x_1)),\ldots,(x_m,c(x_m))\}$, drawn independently and identically distributed (i.i.d.) from D.
- **Error (errD(h))**:
  - The probability that the hypothesis hh disagrees with the target concept cc on a randomly drawn instance: errD(h)=Px∼D[h(x)≠c(x)]
- **PAC Criterion**:
  - A learning algorithm is **PAC-learnable** if, for any $0<\epsilon<1$ (error tolerance) and $0<\delta<1$ (confidence parameter), the algorithm can find a hypothesis $h \in H$ such that: $P(errD(h)\le\epsilon)\ge1-\delta$
  - Here, m (number of training samples) depends polynomially on $1/\epsilon$, $1/\delta$, and the complexity of H.

5.2 Key Elements of PAC Learning

5.2.1 Sample Complexity

- The number of training examples required to ensure the hypothesis hh meets the PAC criterion.
- Depends on:
    - Error tolerance (ϵ).
    - Confidence level (1−δ).
    - Size of the hypothesis class (H).

5.2.2 VC Dimension (Vapnik-Chervonenkis Dimension)

- A measure of the capacity or complexity of the hypothesis class H.
- Higher VC dimension indicates a more expressive hypothesis class but may require more data to achieve PAC learning.

5.2.3 Efficient Learnability

- A concept class {C} is efficiently PAC-learnable if there exists a polynomial-time algorithm that satisfies the PAC criterion.

5.3 Advantages of PAC Learning

- Provides a rigorous foundation for understanding machine learning.
- Helps quantify the trade-off between the size of training data, model complexity, and generalisation error.

5.4 Limitations of PAC Learning

- Assumes i.i.d. data, which may not always hold in real-world scenarios.
- Relies on the existence of a hypothesis h ∈ H close to the true concept cc, which might not always be true.

5.5 Comparison: Formal Model vs PAC Learning

Aspect	Formal Model	PAC Learning
Scope	Provides a general framework for learning tasks.	Focuses on the feasibility of learning under uncertainty.

Aspect	Formal Model	PAC Learning
Goal	Define components of a learning system.	Quantify the conditions for successful learning.
Data Requirements	Depends on the specific problem and algorithm.	Requires i.i.d. samples from a probability distribution.
Complexity Measure	Loss function, hypothesis space complexity.	VC dimension, sample complexity.

- **Formal Model**: Lays the foundation for defining and solving machine learning problems in a structured manner.
- **PAC Learning**: Provides a probabilistic framework to evaluate whether a learning problem is solvable under practical constraints. It is particularly useful for understanding the theoretical limits of generalisation and the relationship between data, model complexity and performance.